ResearchAgent



Research with a wallet.

An agent that answers from free sources first, then buys the articles it is missing from their writers (at most one per round, up to three rounds), **over x402 on the XRP Ledger**, inside a budget you set. The writer is paid only by publishing the article's key on the ledger.

- XRPL Testnet · no real money
- Demo writers are fictional; team-authored notes are labelled · TEST-SGD stands in for RLUSD
- Runs on your machine · no hosted instance

— Progress since SingHacks

From a prototype a human approved, to one line that pays on Testnet

19–20 Sep
SingHacks
Prototype: a human approved every purchase.

10 Oct
AI Tinkerers
Live demo; first Testnet payments 7–9 Oct.

2–3 Nov
v0.1–v0.8
MCP server, Mac app, escrow, hire the agent, audit trail.

4 Nov
rc.1
Install without cloning.

5 Nov
rc.2
Real Testnet without cloning; key reveal by default.

7 Nov
v1.0.1
`try`, TEST-SGD escrow, team notes, Settings.

8 Nov
v1.0.2
Anchor-pending fix, anchored abstentions in the app, honest plan text.

12 Nov
Finals
Final acceptance, then the pitch.

183
pull requests merged since SingHacks (21 Sep–7 Nov)

13
tagged releases, v0.1.0 → v1.0.2

1,369
TypeScript tests at post-#398 integration; 258 Python tests (#393)

224
shared receipt-conformance vectors; the TypeScript and Python verifiers agree on all

gh pr list --state merged · gh release list · PRs #381 (vectors), #393 (Python), #396 (post-#398 TypeScript count) · docs/SFF-PUSH.md

One line, no clone: the whole judge path on real Testnet

```
$ npx -y …/releases/download/v1.0.2/researchagent-mcp-1.0.2.tgz try --testnet
```

```
Testnet · no real value · Fixture models and fictional writers
1. Create 9126E19D…B015B8CD
2. Finish · sha256(K) = Condition ✓ 805AA52F…1342EBB6
3. Refund · writer-signed BFAC5EF0…020585D1
4. Re-buy create 278087AB…0749871B
5. Re-buy finish ✓ 484A818A…AEB4C2EC8
6. Anchor 22081E19…A1E922F9
PASS testnet 12/12 (read-only, s.altnet)
RESULT PASS
Finished in 82.0 s · budget S$2.00 · spent S$0.70
```

Fresh npm cache, one Mac	rc.2	Now
try: install → six links → RESULT PASS	—	**84.2 s**
MCP package	~~21.4 MB~~	**3.4 MB**
MCP cold initialize	~~56.5 s~~	**6.3 s**
Local web app ready	~~134 s~~	**27.0 s**

> Measured 7 Nov with fresh faucet wallets. **Two of four cold attempts beat the 90 s deadline** (73.8 s, 84.2 s); two overran. If the faucet fails, it prints a labelled SIMULATED run and its reason.

PR #387 · docs/evidence/round4-j2/ (try-output.txt, try-timing.json, cold-timings.json, try-timeout*.json) · before = round-4 mock-judge observations on rc.2

— What we built

One question, one budget, and only code can spend it

1

Ask

A question and a budget, S$0 to S$5.

2

Confirm the plan

A 5-second sheet. It confirms the plan, not a purchase.

3

Search writers

Each writer returns abstracts and a signed fact menu, never premium text.

4

Decide

A decision model judges what is worth buying, weighted by trust.

5

Pay

Policy code locks payment in a key-reveal escrow on XRPL, within S$1 per source.

6

Verify, answer, anchor

Decrypt, check every promise, refund a broken one, cite, anchor the audit head.

Five hard gates, on every run

1. No premium bytes before a grant.
2. The budget is the only spending authority; only policy code can buy, never the LLM.
3. One charge per intent, across retries and restarts.
4. Every citation points at a delivered passage.
5. Anything simulated or substituted is labelled.

0 violations

Read-only gate audit: 10 runs, 568 observations, 48 fresh questions (3 Nov). Chaos: 200 runs with real SIGKILLs, 0 double charges (4 Nov). MCP spending caps default to zero; the calling model cannot raise them.

— Creativity · 20%

The writer is paid only by publishing the key: delivery and payment are one ledger event

Writer

Encrypt and promise

Fresh key K per buyer. Signs the quote: `sha256(K)`, ciphertext hash, passage manifest, price.

Buyer policy

EscrowCreate

Native XRPL Escrow. Condition = `sha256(K)`, with a CancelAfter deadline.

Writer

EscrowFinish

To be paid, the writer must reveal K as the fulfilment. K is now public on the ledger.

Buyer policy

Decrypt and check

AES-256-GCM decrypt, check every promised passage, then grant and cite.

Promise kept	Create	→ Finish, K revealed	→ decrypt ✓ passages ✓	→ cited	`278087AB…` →	`484A818A…`
Promise broken	Create	→ Finish, K revealed	→ claim missing	→ writer-signed refund + fraud proof	`805AA52F…` →	`BFAC5EF0…`
Writer no-show	Create	→ no key by CancelAfter	→ EscrowCancel, principal back, nothing granted		`8DA4ECE7…` →	`F89DD9B1…`

> **What the ledger does and doesn't enforce.** Key reveal makes **delivery** atomic: no key, no payment. **Quality** rests on the signed promises, a fraud proof anyone can check, and reputation. After a finish, the refund is writer-signed: our fixture writer cooperates, and a real writer could refuse. No smart contract: Escrow with a PREIMAGE-SHA-256 condition is native XRPL.

The same key-reveal escrow, settled in a stablecoin-style token

TEST-SGD through XLS-85 token escrow

- **Opt in:** add `--iou` to the zero-clone MCP or web app. S$1 maps to 1 TEST-SGD.
- **Preflight refuses** a missing issuer locking permission, frozen or unauthorized trust lines, short balances and transfer fees.
- **Both verifiers** check currency, issuer and the writer's trust-line balance change, since EscrowFinish has no delivered_amount: TypeScript and Python PASS 6/6.
- **Testnet runs:** three purchases in 15.0–18.4 s; a no-show cancel returned the principal in 132.3 s.

`create E36AF9F1…18878C5C · finish 384CA34F…0181C02F5 · anchor 8174E3BE…ADC35A3`

Official RLUSD: payments yes, escrow one flag away

- **Payments work.** On 3 Nov, official Testnet RLUSD paid a writer 0.90 (`1F2449A4…`) and a broken promise was refunded 0.30.
- **Escrow is blocked by the issuer, not the network.** TokenEscrow is enabled on Testnet, but the RLUSD issuer's flags (2174353408) lack AllowTrustLineLocking. One AccountSet by the issuer would allow it.
- **So we escrow our own token** and say so on every amount, receipt and ledger row. It is our top builder-feedback item.

TEST-SGD · IOU issued by our Testnet issuer · no real value · stand-in for RLUSD

No peg, backing or mainnet claim

PRs #381, #384 · docs/evidence/m1-iou/, round3-e/runs.json, rlusd.md · BUILDER-FEEDBACK §1 (issuer flags at ledger 22058675; refusal predicted, never submitted)

— Technical depth · 20%

Don’t trust our agent. Verify it, against the ledger.

```
$ pip install …/researchagent-1.0.2…whl
$ researchagent verify UC3.receipts.json --ledger
PASS checks: 57 PASS, 0 FAIL, 2 SKIP
PASS audit anchor: Validated XRPL Testnet ledger 22057921
PASS testnet 12/12 (read-only, s.altnet)
RESULT PASS
```

Python, standard library only, no Node. Any receipt the app exports (Export ⋯ → receipts.json) can be checked the same way.

- **Two independent verifiers**, TypeScript and Python, agree on all 224 shared vectors, including tampered amounts, issuers, refunds and a missing refund.
- **Historical 61/61 Testnet transactions re-validated live** on our public ledger page in 1.65 s (7 Nov), read-only.
- **Every finished Testnet answer is anchored:** an AccountSet whose memo is the audit-chain head, 12 drops, including a run that bought nothing.
- **Persist before submit.** A restart reconciles the same signed hash and never signs twice; a missing transaction is *pending*, not failed.
- **Exact delivery.** Partial payments rejected before submit; delivered amount, currency and issuer checked after.
- **Prompt injection can’t spend:** model inputs are capped and untrusted, and purchases come only from policy code.

PRs #377, #381, #387, #388 · docs/evidence/round4-j2/python-verifier.txt (README UC3 evidence) · round4-l4/live-ledger.json · BUILDER-FEEDBACK top five

Every way in runs from one package, for people and for other agents

One-line judge path

`try --testnet` buys, refunds, re-buys, anchors and verifies, then exits.

84.2 s from an empty cache · six links · RESULT PASS

Claude Code, any MCP client

`claude mcp add … --testnet`. Seven tools; operator caps default to zero.

Cold initialize 6.3 s · `--iou` for TEST-SGD

Local web app, no clone

`app --testnet` serves the real UI on 127.0.0.1 with faucet wallets. Ask your own question.

Ready in 27.0 s from an empty cache

SDKs: TypeScript and Python

Ask, stream, verify. No purchase method: only backend policy can buy.

Python verifier needs no Node

Writers in one command

`publish ./notes --testnet`: faucet wallet, XRPL Credential, key-reveal escrow.

Credential in 43.8 s · first try from `init`: `33.9 s`

Hire the agent over x402

Another agent pays ResearchAgent; it buys, pays a royalty to every cited writer and returns the change.

Five-hop chain on Testnet, 3 Nov

> XRPL questions · fixture decision model · Testnet: team-authored real build notes, not independent. Held-out v4 7/20 bought and cited, 0/10 off-topic purchases; key fact leads 2/20; 0 gate violations. Dev, not held-out: v1 16/20, v3 7/20. Local only: no hosted instance (`{{REGISTRY_STATUS}}`).

— Feasibility · 20%

Fractions of a cent in fees, and it recovers from failure

435 drops

network fees per key-reveal purchase

12 create + 423 conditional finish, measured on Testnet

S$0.00087

fees per purchase, projected

At an assumed S$2.00 per XRP; a projection, not a quote

US$0.0026

model cost per question

Live decision and coverage models, 48 fresh questions, 3 Nov

894 drops

all six UC3 transactions

Two buys, a refund and an anchor; anchor alone 12 drops

43.9 s

nightly Testnet purchase in CI

Devcontainer boots and buys on real Testnet; green 6 and 7 Nov

0

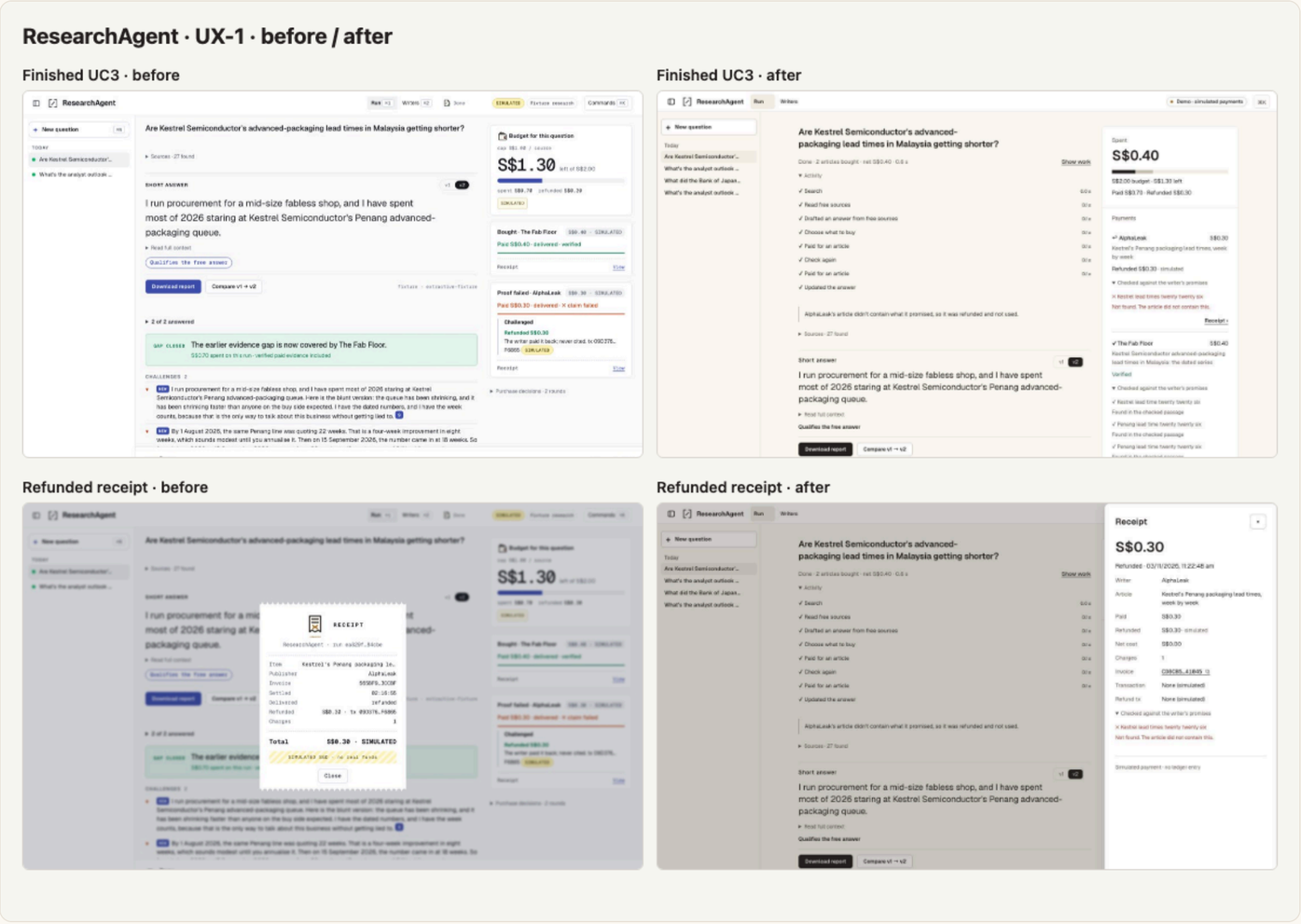
double charges under chaos

200 runs with real SIGKILLs; reconciler recovers in-flight payments

> **Does it buy the right things?** On 48 fresh synthetic questions, **93.15%** of gross article spend bought a requested fact retained in the final cited answer (95% CI 81.25–100.00%); wasted spend 6.85% (0.00–18.75%). Live decision models, fixture research, simulated payments, 3 Nov. The gain over the previous policy (+3.08 pts, 0.00–9.43) is not significant; the labels are same-vendor and await human review.

README cost card (round3-a) · round4-l4/judge-path.md · eval/scorecard/fresh-results.json · FRESH-AUDIT.md · codespace-testnet.yml · PRs #376, #381

The money is the hero, and every label tells the truth



- **One environment indicator** in the header: “Testnet · no real value” or “Simulated”.
- **Payment rows change state in place:** escrow created, key published, sha256(K) = Condition ✓, refunded. Each opens its receipt and hash.
- **A refund says why and who signed:** “Refund · writer-signed · fixture writer cooperates (a real writer could refuse).”
- **Proof stays out of the way:** “6/6 Testnet hashes verified live ✓ · show” expands on demand.
- **Motion only for money**; the pixel agent marks loading states only and never stands in for a receipt.
- **One Settings page** for the web app and the Mac app: nine sections; changes to caps and keys are guarded and audited.

Image: release v0.5.0 before-after.png (offline demo, simulated payments) · PRs #379, #385, #386, #388 · docs/design/UX-DIRECTION.md

Ranked builder feedback for Ripple, with runnable repros

1 Issuer permission blocks RLUSD escrow

TokenEscrow is enabled, but the issuer lacks AllowTrustLineLocking. Fix: a network and issuer preflight, with dated flags.

2 An issued amount must mean the delivered amount

Amount alone can’t prove exact delivery, and EscrowFinish has no delivered_amount. Fix: negative issued-asset vectors.

3 Public endpoints disagree on retained history

Unavailable history must not read as failed settlement; HTTP lacks browser CORS. Fix: document retention and archival lookups.

4 Escrow expiry doesn’t return funds by itself

A failed-proof escrow stayed locked past its deadline until an explicit EscrowCancel. Fix: document expiry vs cancellation.

5 New: a signed IOU quote can overflow the 1 KiB Memo

Observed 7 Nov on the TEST-SGD path. Fix: check serialized Memo size before signing and return the reason.

+ Mainnet readiness: a gap checklist, not a claim

Key custody (KMS or external signer), FX quotes, facilitator SLA, KYB and sanctions screening.

> **7** ranked findings with repro scripts · **33** numbered findings (B01–B33) · **15** September submissions · **7** upstream issue drafts with a filing kit. Filed upstream: `{{UPSTREAM_FILED_URLS}}`.

— Check it yourself

Every claim has a Testnet hash or a command

What	Where
Run the judge path	`npx -y … try --testnet`
Release, every artifact	`v1.0.2` on GitHub Releases
Replay of a Testnet run	…/theFastandtheFungible/demo/
61 hashes, checked on load	…/theFastandtheFungible/ledger/
Demo video	`{{VIDEO_URL}}`
Code and evidence	github.com/Collaboration95/…

Before mainnet

- Payer keys in a KMS or external signer
- RLUSD escrow once the issuer allows it; live FX
- KYB and a real sanctions-screening provider
- Independent writers with consent (none yet)
- Signed, notarised Mac app

